WHAT SURVIVES THE ETH REFUTATIONS? A MAP OF FINE-GRAINED HYPOTHESES AND CONDITIONAL LOWER BOUNDS

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Abstract. Many lower bounds in exact, parameterised and fine-grained complexity are proved from the Exponential Time Hypothesis (ETH) or the Strong ETH. In October 2026 OpenAI and Anthropic claimed deterministic subexponential-time algorithms for #k-SAT that refute both. However, a refuted hypothesis does not settle the theorems proved from it: what falls depends on the premise that each proof actually uses, and outside learning theory we know of no systematic account. We present a map of 34 hypotheses and 27 published conditional theorems, assuming the claimed algorithms are correct. Each row is refuted, has lost its premise, or survives; a fourth label marks the few theorems whose conclusion is now contradicted by an explicit algorithm. The map rests on one distinction. The new algorithms handle k-CNF formulas of bounded width and, through the width reduction of Calabro, Impagliazzo and Paturi, CNFs of any width with at most $n \cdot 2^{o((\log n)^{1/4}/\sqrt{\log\log n})}$ clauses (our derivation from the explicit rate of the Anthropic manuscript). Nothing is known for CNF-SAT with polynomially many clauses of unbounded width. Many of the best-known lower bounds (treewidth optimality, edit distance, Fréchet distance, LCS, diameter approximation) are proved from that "CNF-SETH" and survive with their premise intact. Gap-ETH and the ETH-based proof of the Parameterized Inapproximability Hypothesis lose their force; Almost-ETH, OV and the Set Cover Conjecture are untouched. We list the few new algorithms that follow and place the first follow-ups: crossover points, barriers, and the Satisfiability Hypothesis (SH). A refutation removes premises; it does not by itself give algorithms, and nothing here is practical. Everything is conditional on the claimed refutations, still under review. Supporting notes are in our public repository, https://github.com/xennygrimmato/eth-hamiltonicity.

1. Introduction

A large part of exact, parameterised and fine-grained complexity rests on two hypotheses. The Exponential Time Hypothesis (ETH) of Impagliazzo and Paturi [42, 43] defines s_k, for $k \geq 3$, as the infimum of the δ such that k-SAT on n variables is solvable in time $O(2^{\delta n})$, and asserts $s_k > 0$ for $k \geq 3$. The Strong ETH (SETH) is $s_\infty = \lim_k s_k = 1$ [43]. Lower bounds proved from them range from $2^{o(n)}$ bounds for NP-hard graph problems [44, 55] to quadratic bounds for problems in P, such as edit distance [9].

1.1. The two refutations.

In October 2026 two manuscripts appeared, by OpenAI [68] and by Anthropic [24]. Each claims a deterministic algorithm that counts the satisfying assignments of a k-CNF in subexponential time. Both come with Lean 4 formalisations of their top-level statements. These are *claimed* results, still under review. We write [A] and [O] for them, and use them in the following forms.

Imported Theorem 1.1 ([A], Main Theorem, eq. (1))**.** *There is an absolute constant C and a deterministic algorithm that counts the models of a k-CNF with n variables and m clauses in time*

$$2^{C\cdot\min\{k^2 n\log\log n/\sqrt{\log n},\; km/\sqrt{\log m}\}}.$$

Imported Theorem 1.2 ([O], main theorem)**.** *For each fixed k there is a deterministic algorithm that counts the models of a k-CNF with n variables and m clauses in $(n+m)^{O_k(1)}2^{O_k(n\log\log n/\sqrt{\log n})}$ bit operations and polynomial space, and in $2^{O_{k,D}(n/\sqrt{\log n})}$ when $m \leq Dn$.*

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Two features matter below. First, Imported Theorem 1.1 is *uniform in k*: C is absolute, so the bound is subexponential in n as long as $k = o((\log n)^{1/4}/\sqrt{\log\log n})$. [O] states no dependence on k, so it is not known to be uniform. Uniformity is irrelevant for ETH and SETH. Each engine gives $s_k = 0$ for every fixed k, hence $s_\infty = 0$ ([A], Corollary 5; the reduction from 3-SAT to k-SAT is the deterministic sparsification of [44]). It matters only for formulas whose width grows with n (Section 2). Second, both algorithms are deterministic and count. So they refute the randomized, counting and nondeterministic variants of ETH and SETH *for k-CNFs* at once ([A], Corollaries 5 and 6 and the remark after Corollary 5).

1.2. **The problem: which results fall?** However, a refutation of ETH does not tell us which of the results proved from it fall. A conditional lower bound "$H \Rightarrow$ problem X needs time T" is proved by a reduction *from* a problem covered by H *to* X. Refuting H removes the premise; it says nothing about X. Worse, the premise named in a theorem is often stronger than the premise its proof uses. OpenAI [68, §1.1.3] already observe this for several bounds: those for Fréchet distance, diameter approximation and the treewidth dependence of Independent Set and Dominating Set also follow from CNF-SAT of unbounded width, which their theorem does not reach, and new algorithms need reductions in the other direction. For learning theory, Li [50, §6] takes stock of the lower bounds and of the fine-grained hypotheses they use. Beyond this and such remarks we are unaware of a systematic account of what the refutations affect across areas, and without one every result stated "under ETH" or "under SETH" is in doubt.

We consider the *classification problem*: for each fine-grained hypothesis and each published theorem $H \Rightarrow X$, decide what follows from [A] and [O]. We take the view that the status of a theorem depends on its premise *as actually used in the proof*, not as stated. Throughout, "refuted" means *refuted assuming the claimed refutations*, and we use the following statuses.

- **refuted**: the hypothesis itself is false.
- **premise refuted**: a theorem $H \Rightarrow X$ with H false; nothing here contradicts X.
- **survives**: we know of no consequence of the refutations that touches the statement. For a theorem this means that its premise, as actually used in the proof, still stands.

The additional label **conclusion refuted** marks a theorem $H \Rightarrow X$ whose conclusion X is contradicted by an explicit algorithm. One row is a theorem whose premise is the *negation* of ETH, so that the theorem now **fires** (Section 4.1). A few rows are **mixed** and are split in the reason column. Statuses record what follows from the two ETH refutations only. Where a row has since been claimed refuted on unrelated grounds, the reason column says so (A23–A25).

Example 1.3 (Running example)**.** Lokshtanov, Marx and Saurabh [54] prove that Independent Set has no $(2-\varepsilon)^{\mathrm{tw}}$ algorithm, under SETH. By the classical argument [44, 55], ETH implies that Independent Set on n vertices has no $2^{o(n)}$ algorithm. Both theorems cite a refuted hypothesis, yet they end up in different places. The second has its conclusion refuted: [A], Corollary 7, gives $O(2^{\delta n})$ for every $\delta > 0$ (row B13). The first survives: its proof reduces from CNF-SAT with m clauses of unbounded width, which the new algorithms do not reach (row B1, Section 2). We return to this pair in Section 6.

1.3. **The naive route and the principle.** At first one might expect every SETH-based bound to fall with SETH, and every ETH-based bound with ETH. However, Example 1.3 shows that this is wrong in both directions. A bound can survive because its proof never needs bounded width, and a refutation can contradict a conclusion only through an algorithm, never through the reduction that proved the bound. Instead, we rely on two rules. A row falls only if the premise its proof uses falls. And a row gains an algorithm only if some reduction runs in the other direction: from X to k-SAT with a linear number of variables (the SERF framework of [44], and [A], Corollary 7), or from more general theorems inside the manuscripts that the top-level statements do not expose (our own sparse results, Section 5). Most rows therefore lose a premise and gain nothing. We say so row by row.

1.4. **Contributions.** Our contributions are:

- We separate bounded from unbounded width. We prove that Imported Theorem 1.1 counts the models of CNFs of any width with at most $n \cdot 2^{o(L(n))}$ clauses in time $2^{o(n)}$, where $L(n) = (\log n)^{1/4}/\sqrt{\log\log n}$, and that nothing follows at $n^{1+\Omega(1)}$ clauses (Section 2).
- We classify 34 hypotheses: 13 are refuted, 20 survive and 1 is mixed (Section 3).

- We classify 27 conditional theorems: 3 have their conclusion refuted, 1 fires, 10 lose their premise, 9 survive and 4 are mixed (Section 4).
- We list the new algorithms that follow, and separate those that need only the formalised top-level theorems from those that need internal results of [A] (Section 5).
- We read the map by area, and place the first follow-up works by kind: new algorithms, constants and crossover points, barriers, and replacement hypotheses (Sections 6 and 7).

Much of the classification is bookkeeping that an expert in each area could do. What we add is one map across the areas, with each row checked against its primary source where one exists.

1.5. **Sources and status of each claim.** The map was compiled from the primary sources. For every row tagged ^{L} in the lists of Sections 3 and 4 we checked the wording against the abstract or full text of the cited source (arXiv, ECCC, DROPS, journal, or authors' copy). Rows tagged ^{F} are folklore conjectures without a single primary statement; ^{I} marks our own inference, and ^{O} our own results. In an earlier pass, nine rows rested on secondary sources or on sources we had not re-read (ETH itself; the CIP duality; Williams' Max-2-CSP algorithm; Feige's hypothesis; Pătraşcu–Williams, twice; Cai–Juedes; Chen–Huang–Kanj–Xia; Marx's treewidth paper). We re-verified all nine against the primary text or abstract for this note, so no row now rests on a secondary source. Theorem numbers refer to the versions named in the bibliography.

The working map behind this note, `research/fine-grained-map.md`, names for each row the source we checked, with its URL where there is one. It is in our public repository `https://github.com/xennygrimmato/eth-hamiltonicity` [82], together with the Lean proofs of our own results that we cite. Rows A3, A23–A25, A33, A34, B13–B17, B22–B24 and B27 of this note correct or extend the map at the commit we cite: they add priority credits, add the follow-up works found by a novelty search of arXiv and Hexagon on 5 November 2026, and split B16 by what is machine-checked. We separate three kinds of claim. *Proved on paper*: Proposition 2.1 and the observations of Section 5. *Machine-checked in Lean*: only the results of ours that Section 5 labels so; each uses only the standard axioms and is built on the labs' own Lean developments. *Conjectured, not claimed*: the further sparse problems of Section 5. Everything rests on [A] and [O].

Outline. Section 2 separates bounded from unbounded width. Sections 3 and 4 give the two tables. Section 5 lists the new algorithms, Section 6 reads the map by area, and Sections 7 to 9 give related work, a summary and the limitations.

2. Bounded width falls, unbounded width stands

In this section we state three facts used throughout, derive the density frontier of the new algorithms, and name the hypothesis that survives.

2.1. **Three facts.**

(F1) $s_k = 0$ for every k. This refutes ETH, SETH, #ETH, #SETH, ⊕SETH and NSETH in their k-CNF forms.

(F2) *Local witnesses.* A problem with complexity parameter n whose yes-instances have an $O(n)$-bit witness checkable by an $O(n)$-gate circuit, or by bounded-width clauses on $O(n)$ variables, gets deterministic time $2^{\delta n}$ for every $\delta > 0$ by Tseitin's transformation. Counts follow when the encoding is parsimonious. This is the SERF route [44]; [A], Corollary 7, uses it with a cardinality circuit of $O(n)$ gates.

(F3) *The width frontier* (Proposition 2.1 below). CNFs of arbitrary width are reached only when the number of clauses is barely superlinear.

2.2. **The width reduction and the density frontier.** Calabro, Impagliazzo and Paturi [18] relate CNF-SAT of bounded density to k-SAT by Schuler's width reduction. While some clause C has more than k literals, pick k of them, C', and branch on "C' is satisfied" (replace C by C') versus "C' is falsified" (set its k literals to false). The branches are disjoint, so model counts add up, and the leaves are k-CNFs. This counting form is [81, Lemma 6.1]. With constant k it gives #CNF-SAT with $m \le cn$ clauses in time $2^{\varepsilon n}$ for all $c, \varepsilon > 0$, from either engine [81, Corollary 6.2]. The uniformity of Imported Theorem 1.1 lets k grow, and gives a little more.

Proposition 2.1 (our derivation from Imported Theorem 1.1)**.** *Put $L(n) = (\log n)^{1/4}/\sqrt{\log\log n}$. Assume Imported Theorem 1.1. For every function f with $f(n) = o(L(n))$ there is a deterministic algorithm that counts the models of CNF formulas of arbitrary width with n variables and at most $n \cdot 2^{f(n)}$ clauses in time $2^{o(n)}$.*

Proof. Let $m \le Dn$ with $\log D = f(n)$, and fix $k \ge 2$, to be chosen. Run the width reduction. Along a root-to-leaf path every "satisfied" branch removes one long clause, so there are at most m of them, and every "falsified" branch fixes k variables, so there are $r \le n/k$ of them. The leaves with exactly r falsified branches number at most $\binom{m+r}{r}$ [18, proof of Lemma 5], so there are at most $(n/k+1)\binom{m+\lfloor n/k\rfloor}{\lfloor n/k\rfloor} \le (n+1)\big(e(Dk+1)\big)^{n/k}$ leaves. Each leaf is a k-CNF with $n' \le n$ variables and at most m clauses. If $n' \le \sqrt{n}$, count by brute force in time $2^{\sqrt{n}}\,\mathrm{poly}(m)$. Otherwise $\log n' \ge \frac{1}{2}\log n$, so $\log\log n'/\sqrt{\log n'} \le 2\log\log n/\sqrt{\log n}$ for n large, and Imported Theorem 1.1 counts the leaf in time 2^{sn} with $s = 2Ck^2\log\log n/\sqrt{\log n}$. The total time is at most

$$\mathrm{poly}(n,m)\cdot 2^{\,(n/k)\,(f(n)+\log(k+1)+2)\,+\,sn\,+\,\sqrt{n}}.$$

Take $k = \lceil \sqrt{\max\{f(n),1\}\cdot L(n)}\,\rceil$. Then $k \to \infty$, $f(n)/k \le \sqrt{f(n)/L(n)} \to 0$ and $(\log k)/k \to 0$, while $k/L(n) \to 0$ gives $s = 2C(k/L(n))^2 \to 0$. Every term of the exponent is $o(n)$, and $\mathrm{poly}(n,m) = 2^{o(n)}$ since $m \le n^{1+o(1)}$. □

Remark 2.2 (what does not follow).

(a) The clause-count branch of Imported Theorem 1.1 does worse here: it needs $km = o(n\sqrt{\log m})$, which allows only $m \le n(\log n)^{1/2-o(1)}$.

(b) Through [O] only constant density is reached explicitly, since its O_k-constants are not stated; a diagonal argument gives some unbounded but non-explicit density.

(c) For $m = n^{1+\Omega(1)}$, or $m \ge n\cdot 2^{(\log n)^{1/4+\Omega(1)}}$, nothing follows: the leaf count forces $k \gtrsim \log(m/n)$, where neither engine is subexponential in n. Hence the statement

> *CNF-SETH*: for every $\varepsilon > 0$, CNF-SAT with n variables and m clauses of unbounded width is not solvable in time $2^{(1-\varepsilon)n}\,\mathrm{poly}(m)$

survives, together with its counting, parity and nondeterministic forms.

CNF-SETH is exactly the premise that many famous reductions use. Lokshtanov, Marx and Saurabh [54] assume that "SAT cannot be solved in $(2-\varepsilon)^n m^{O(1)}$ time" and stress that they need no sparsification. Backurs and Indyk [9] reduce from CNF-SAT with M clauses to edit distance. Bringmann's "SETH′" [14] is "no $O^*((2-\delta)^N)$ algorithm for CNF-SAT". Roditty and Vassilevska Williams [70, Theorem 1] conclude "CNF-SAT on n variables is in $O^*((2-\delta)^n)$". And the moderate-dimension Orthogonal Vectors conjecture follows from CNF-SETH by Williams' split-and-list reduction with $d = m$. OpenAI [68, §1.1.3] make the same observation for the Fréchet, diameter and treewidth bounds, and note that the LCS bounds follow from OVC. What dies is everything that genuinely goes through sparsification to $O(n)$ clauses, or through bounded width.

3. Hypotheses: thirteen refuted, twenty survive

In this section we classify the hypotheses. The list below gives them. "Gives algorithms" is noted in the reason column when a refutation also yields an explicit faster algorithm for the problem named.
Hypotheses, assuming the claimed refutations. Each row gives the hypothesis and its source, its status, and the reason.

A1 ETH: $s_k > 0$ for $k \ge 3$ [42, 43]^{L}. *Status:* refuted. Either engine; 3-SAT and k-SAT in $2^{o(n)}$.

A2 SETH: $s_\infty = 1$ [43]; randomized algorithms allowed in [27]^{L}. *Status:* refuted. (F1); [O] suffices despite non-uniformity in k.

A3 Density-SETH: CNF of any width with $\le cn$ clauses needs $2^{(1-\varepsilon)n}$ [18]; [63, Conj. 2]^{L}. *Status:* refuted. Counting width reduction [81, Cor. 6.2]; also [50, Prop. 6.1].

A4 CNF-SETH: CNF-SAT $\notin 2^{(1-\varepsilon)n}\,\mathrm{poly}(m)$, unbounded width [54, 9, 14, 70]^{L}. *Status:* survives. Proposition 2.1 and Remark 2.2: reach ends at $n2^{o(L(n))}$ clauses.

A5 Super-SETH: for every unbounded f, k-SAT $\notin 2^{n(1-f(k)/k)}$ [83]^{L}. *Status:* refuted. It implies SETH.

A6 #ETH [33, p. 21:5]^{L}. *Status:* refuted. [A], Corollary 6; [O]. Gives #3-SAT in $2^{o(n)}$.

A7 #SETH, ⊕SETH for k-CNF; equivalent to k-⊕Hitting Set and k-⊕Set Cover hardness [27, Thm. 1.3]^{L}. *Status:* refuted. (F1).

A8 #CNF-SETH, ⊕CNF-SETH (poly(m) clauses)^{I}. *Status:* survives. As A4.

A9 NSETH: $\forall\varepsilon\,\exists k$, k-TAUT $\notin$ NTIME$[2^{(1-\varepsilon)n}]$ [19]^{L}. *Status:* refuted. A deterministic algorithm is a nondeterministic one ([A], after Corollary 5).

A10 NSETH for CNF with poly(m) clauses, or for poly-size circuits [63, Conj. 3]^{L}. *Status:* survives. As A4.

A11 MA-SETH, AM-SETH. *Status:* refuted (2016). Williams [84]: MA proofs in c^n time, $c < 2$, for #SAT of sublinear-depth circuits; "strongly refute" both^{L}. For k-CNF also by (F1).

A12 Gap-ETH: no $2^{o(n)}$ (randomized) algorithm distinguishes satisfiable 3-CNFs from those not $(1-\varepsilon)$-satisfiable [34, 60], as quoted in [20]^{L}. *Status:* refuted. An exact deterministic $2^{o(n)}$ decider distinguishes the cases, as [O] notes (§1.1.1). [59] also records Gap-ETH as false.

A13 ETH for Circuit-SAT of linear size; VSP-Circuit-SAT of size cn [27, Thm. 1.1(v)]^{L}. *Status:* refuted. Tseitin gives $O(n)$ variables; gives algorithms.

A14 Circuit-, NC- and BP-SETH [4]^{L}. *Status:* survives. Superlinear-size representations.

A15 FPT $\neq$ M[1] (miniaturised CNF of size $k \log n$) [36, §3.5]. Chen–Grohe: M[1] = FPT iff 3-SAT $\in$ DTIME$(2^{o^{\mathrm{eff}}(n)})$ [23]^{L}. *Status:* refuted. The iff needs an effective little-o; [A] (rate with absolute C) and [O] at $k = 3$ supply one. Gives algorithms for M[1]-complete problems.

A16 FPT $\neq$ W[1]. *Status:* survives. ETH implies it, not conversely; [23, §6] list "ETH $\Leftrightarrow$ FPT $\neq$ W[1]?" as open^{L}.

A17 Low-dimension OVC: $\forall\varepsilon\,\exists c$, OV with $d = c \log n \notin n^{2-\varepsilon}$ ("LDOVC" of [38])^{L}. *Status:* survives. SETH support gone (it was via sparsification [15, Lemma 2.1]); [A] notes it does not refute OV.

A18 Moderate-dimension OVC: no $n^{2-\varepsilon}\,\mathrm{poly}(d)$ [38]^{L}. *Status:* survives. Implied by CNF-SETH with $d = m$ [2, Lemma 1]; premise intact.

A19 k-OV, low and moderate dimension^{I}. *Status:* survives. As A17, A18.

A20 Hitting Set conjecture [8]^{L}. *Status:* survives. Not known to follow from SETH; implies OVC.

A21 Set Cover Conjecture: $\forall\varepsilon\,\exists k$, Set Cover with sets of size $\leq k \notin 2^{\varepsilon n}\,\mathrm{poly}(m)$, n the universe [27, Thm. 1.2]^{L}. *Status:* survives. Not implied by SETH even before 2026; the natural witness has $\Theta(n \log n)$ bits, so (F2) does not apply.

A22 k-Hitting Set, k-Set Splitting, k-NAE-SAT need $2^{(1-\varepsilon)n}$ for large k, each equivalent to SETH [27, Thm. 1.1]^{L}. *Status:* refuted. Equivalent to SETH; directly, they are k-CNFs on the universe. Gives algorithms.

A23 3SUM conjecture^{F}. *Status:* survives. No SAT premise; see B26. Claimed refuted independently of SAT, in $O(n^{1.9992})$, by Alman and Vassilevska Williams [6]^{L}.

A24 APSP; combinatorial BMM; OMv^{F}. *Status:* survives. Unrelated to SAT. APSP claimed refuted independently, in $O(n^{2.9995})$ [6]^{L}.

A25 k-Clique in $n^{\omega k/3-\varepsilon}$; min-, exact- and zero-weight k-Clique^{F}. *Status:* survives. Unrelated to SAT. Zero-weight k-Clique claimed refuted independently [6]^{L}.

A26 (ℓ, k)-Hyperclique hypothesis [53, Hyp. 1.4]^{L}. *Status:* survives. Its refutation would give dense Max-k-SAT in $2^{(1-\varepsilon)n}$, which A27 leaves open; no converse from SAT.

A27 Dense Max-3-SAT $\notin 2^{(1-\varepsilon)n}$; Max-2-SAT, Max-Cut not below $2^{\omega n/3}$ (algorithm: [85])^{F}. *Status:* survives. A counter over $\Theta(n^2)$ clauses has $\Theta(n^2)$ variables.

A28 Subset Sum $\notin 2^{(1/2-\varepsilon)n\,\mathrm{F}}$. *Status:* survives. Weights are not local.

A29 Basic QSETH (CNF-SAT) [16]; its k-CNF variant^{L}. *Status:* mixed. CNF form survives (as A4); k-CNF variant refuted [81, §6.2, entry 23].

A30 PIH [57]^{L}. *Status:* survives. Unproved again: both proofs, from Gap-ETH and from ETH [39], lose their premises.

A31 ETH for End-of-a-Line [71]^{L}. *Status:* survives. A different, query-type hypothesis.

A32 Feige's random 3-SAT hypothesis [37]^{L}. *Status:* survives. About polynomial-time average-case refutation.

A33 Almost-ETH: $\forall\gamma > 0$, 3-SAT $\notin O(2^{n^{1-\gamma}})$ [59]^{L}. *Status:* survives. Both engines give $2^{n/(\log n)^{\Theta(1)}} = 2^{n^{1-o(1)}}$, not $2^{n^{1-\gamma}}$. Called Weak ETH in [62], which proves it equivalent to the Satisfiability Hypothesis (SH) of Stearns and Hunt (CNF-SAT of input length L is not in $O(2^{L^{1-\varepsilon}})$), and #SH equivalent to Weak #ETH [62, Thm. 2.4]^{L}.

A34 Sparse Max-SAT: for every $\varepsilon > 0$ some c makes Max-3-SAT on cn clauses need $2^{(1-\varepsilon)n}$, and Max-2-SAT need $2^{(\omega/3-\varepsilon)n}$ [5, Hyp. 3, 4]^{L}. *Status:* refuted. Exact Max-k-SAT with $O(n)$ clauses in $2^{o(n)}$ (B15); recorded by Li [50, Cor. 4.5] and Majdoddin [58, §3.8]. Gives algorithms.

Remarks. Row A11 was refuted before 2026 and is listed for completeness. Row A15 is the one place where an *effective* rate matters. Chen and Grohe's miniaturisation isomorphism maps SUBEPT to FPT, and the explicit rates of both manuscripts make the conclusion M[1] = FPT unconditional modulo the refutations. Manurangsi proposed row A33 in the 27 October 2026 revision of [59], because of the refutations; its comment states that the earlier versions' ETH and Gap-ETH premises "are now known to be false". Row A34 was posed because no sparsification lemma is known for Max-SAT [5]; it falls because its sparse instances are local in the sense of (F2). Two further proposals concern what should replace ETH. The MIT Hardness Group [62] argue for SH, which is equivalent to Almost-ETH (A33), and prove a transfer theorem: a polynomial-time reduction from 3-SAT with s variables and clauses to instances of size $N \le s^{b+o(1)}$ rules out time $O(2^{N^{1/b-\varepsilon}})$ under SH [62, Thm. 3.1]. Li [50, Hyp. 2.14] proposes a length form of ETH at rate C, that CNF-SAT of length L has no randomized $2^{L/(\log L)^C}$ algorithm; as he notes, the new algorithms refute it for $C < 1/2$ and leave it open for $C \ge 1/2$. We mention it here rather than as a row.

4. Conditional theorems: three contradicted, one fires

In this section we classify the conditional theorems (the list below) and then treat the one row whose premise is the negation of ETH, the circuit lower bound of Williams (Section 4.1).
Conditional theorems "$H \Rightarrow X$", assuming the claimed refutations. Each row gives the theorem and its source, its status, and the reason.

Parameterised and exact exponential.

- B1 Treewidth DP optimality: no $(2-\varepsilon)^{\mathrm{tw}}$ IS, $(3-\varepsilon)^{\mathrm{tw}}$ DomSet, $(2-\varepsilon)^{\mathrm{tw}}$ Max-Cut, $(3-\varepsilon)^{\mathrm{tw}}$ OCT, $(q-\varepsilon)^{\mathrm{tw}}$ q-Col, $(2-\varepsilon)^{\mathrm{tw}}$ Partition into Triangles [54]^{L}. *Status:* survives. The premise as proved is CNF-SETH (A4), with $m^{O(1)}$ clauses; no sparsification is used. No faster DP follows.
- B2 Cut&Count pathwidth bounds: no $(3-\varepsilon)^{\mathrm{pw}}$ CVC, FVS, Steiner Tree; no $(4-\varepsilon)^{\mathrm{pw}}$ CDS, CFVS, COCT; Exact k-Leaf Spanning Tree [29, Thm. 1.3, App. D]^{L}. *Status:* survives. Every reduction of App. D starts from CNF-SAT with n variables and m clauses: CNF-SETH. (Their Thm. 1.4 is in B21.)
- B3 Pătraşcu–Williams: k-DomSet in $O(n^{k-\varepsilon})$ for some $k \ge 3 \Rightarrow$ CNF-SAT in $2^{\delta n}\,\mathrm{poly}(m)$, $\delta < 1$ [69, Thm. 2.1]^{L}. *Status:* survives. CNF form.
- B4 Pătraşcu–Williams: d-SUM on N numbers of $O(d \log N)$ bits, $d < N^{0.99}$, in $N^{o(d)} \Rightarrow k$-SAT in $2^{o(n)}$ [69, §5]^{L}. *Status:* premise refuted. Conclusion of the contrapositive is now true; d-SUM bound open.
- B13 ETH $\Rightarrow$ no $2^{o(n)}$ IS, Clique, VC, k-Col, DomSet on n vertices, any density [44, 55]^{L}. *Status:* conclusion refuted. [A], Corollary 7: deterministic $O(2^{\delta n})$ for every δ, any number of edges, with counts by size and of k-colourings. Counting independent sets and q-colourings is also [O], Corollary 1.3. Under SH (A33), no $2^{O(n^{1-\varepsilon})}L^{O(1)}$ for VC, IS and 3-Colouring [62, Table 1].
- B14 ETH $\Rightarrow$ k-VC $\notin 2^{o(k)}\,\mathrm{poly}(n)$ (from B13, as $k \le n$; cf. [17])^{L}. *Status:* conclusion refuted^{I}. $2k$-vertex LP kernel [28, §2.5], then Corollary 7 (Section 5); obtained earlier by Bhattacharjee [10, Thm. 6.13]; we arrived at it independently. Cai–Juedes: Max-SNP-complete problems have $2^{o(k)}p(n)$ algorithms iff 3-SAT $\in$ DTIME$(2^{o(n)})$. Under SH, no $2^{O(k^{1-\varepsilon})}L^{O(1)}$ [62, Table 1].
- B15 ETH $\Rightarrow$ no $2^{o(n)}$ Max-Cut, Max-k-SAT with $O(n)$ constraints^{F}. *Status:* conclusion refuted^{I}. Unweighted, or $O(1)$-bit weights; Section 5, and earlier [10, Rem. 6.5, Cor. 6.4]. Polynomial weights: under ($\mathrm{H_A}$), or by the white-box extension of [O] to $\mathbb{Q}$-algebra weights in [10] (not a consequence of the top-level theorem of [O]). The sparse Max-SAT hypotheses fall with it (A34). Dense: A27.
- B16 ETH $\Rightarrow$ no $2^{o(n)}$ HC, HP, TSP, Steiner, FVS, CVC, CDS, Max-Leaf on sparse graphs ([55, Thm. 3.3] for Hamiltonian Path; folklore for the rest). *Status:* mixed^{O}. Conclusion refuted (machine-checked, under ($\mathrm{H_A}$)): HC, HP, TSP and their directed forms [75] (Section 5). Conclusion refuted on paper, assuming Theorem D.1 of [O] (not a consequence of its top-level theorem): FVS at any density, with counts of every size, and Steiner trees on sparse graphs, by Wu and Wang [88]. Premise refuted: CVC, CDS, Max-Leaf, which we only conjecture, with identities checked numerically [82]; and the remaining problems on dense graphs.
- B19 ETH $\Rightarrow$ k-Clique $\notin f(k)n^{o(k)}$ [21, Thm. 5.5]^{L}. *Status:* premise refuted. $2^{\delta n}$ does not contradict it at small k.
- B20 ETH $\Rightarrow$ binary CSP $\notin f(G)n^{o(k/\log k)}$, k the treewidth [61]; #ETH $\Rightarrow$ #k-matching (bipartite) $\notin f(k)n^{o(k/\log k)}$ [26]^{L}. *Status:* premise refuted. Parameter $k \ll n$. On a class of bounded-degree graphs of unbounded treewidth, Nguyen [64] decides binary CSP in $f(G)n^{O(k/\sqrt{\log k})}$ via the network engine of [A], contradicting the literal (unconditional) form of Marx's Conjecture 1.4 ($n^{o(k)}$); the exponent $o(k/\log k)$ of the theorem is not reached.

B21 ETH $\Rightarrow$ no $2^{o(d \log d)}$ Closest String, Distortion, Disjoint Paths by tw, ... [56]; no $2^{o(p \log p)}$ Cycle Packing [29, Thm. 1.4]^{L}. *Status:* premise refuted.

B22 ETH $\Rightarrow$ $2^{\Omega(\sqrt{n})}$ planar bounds [55]; $2^{\Omega(n^{1-1/d})}$ for fat-object graphs [31] and Euclidean TSP [30]^{L}. *Status:* premise refuted. The refutations give only $2^{n/\mathrm{polylog}\, n}$, worse than the upper bounds. Under Almost-ETH, equivalently SH, the reductions give bounds such as no $2^{O(n^{1/2-\varepsilon})}L^{O(1)}$ for planar Hamiltonian Cycle and no $2^{O(n^{1-1/d-\varepsilon})}L^{O(1)}$ for Euclidean TSP in $\mathbb{R}^d$ [62, Thm. 3.1, Table 1]^{L}.

B11 Under the Set Cover Conjecture: Steiner Tree $\notin 2^{\varepsilon t}\,\mathrm{poly}$, CVC $\notin 2^{\varepsilon t}\,\mathrm{poly}$, Set Partitioning $\notin 2^{\varepsilon n}\,\mathrm{poly}(m)$, Subset Sum $\notin t^{\varepsilon}\,\mathrm{poly}(n)$ [27, Thm. 1.2]^{L}. *Status:* survives. A21 intact.

B12 Subset Sum $\notin T^{1-\varepsilon}2^{o(n)}$ [3]^{L}. *Status:* premise refuted. Reduces from k-SAT; the $T^{1-\varepsilon}\,\mathrm{poly}(n)$ form survives via B11.

Approximation.

B23 Gap-ETH $\Rightarrow$ FPT-inapproximability of k-Clique, k-DomSet, Biclique, induced matching, DkS [20]^{L}. *Status:* premise refuted. Partly re-proved under W[1] $\neq$ FPT (survives): constant-factor k-Clique [51], $k^{1/H(k)}$-factor k-Clique [48], $k^{o(1)}$-factor k-Clique [22], any $F(k)$ for k-DomSet [49]. Only the ETH-based versions lose their premise: the $k^{o(1)}$ bound of Lin, Ren, Sun and Wang [52] and KLM's $T(k)n^{o(k)}$. No W[1] re-proof found for the others.

B24 ETH-hardness of approximation: 2-CSP to $|V|^{1-o(1)}$, Directed Steiner Network to $k^{1/4-o(1)}$ [35]; DkS [59]; best Nash [12]; free games [1]^{L}. *Status:* mixed. DkS to $2^{(\log n)^{1-\varepsilon}}$ re-proved under Almost-ETH in [59] (v3), and under SH in [62, §4]: survives. [35, 12, 1]: premise refuted; quasi-polynomial reductions plausibly give weaker bounds under Almost-ETH^{I}.

B25 ETH $\Rightarrow$ PIH [39]; near-optimal $f(k)n^{o(k)}$-type bounds for approximate k-Clique and CSP under ETH [40]^{L}. *Status:* premise refuted. See A30.

Problems in P.

B5 Edit distance $\notin n^{2-\varepsilon}$ [9]; also from NC-SETH [4]^{L}. *Status:* survives. Reduction from CNF-SAT with M clauses: CNF-SETH.

B6 LCS, DTW: Most-OV with threshold, $n^{2-\varepsilon}\,\mathrm{poly}(d) \Rightarrow k$-LCS $\notin n^{k-\varepsilon}$ [2, Thm. 1, Conj. 1]; binary LCS, edit distance with costs, DTW [15]^{L}. *Status:* survives. [2, Lemma 1]: Conj. 1 follows from Max-CNF-SAT hardness with $\mathrm{poly}(m)$ clauses. [15] is stated under k-SETH but proved from OVH, which CNF-SETH implies.

B7 Fréchet distance $\notin n^{2-\varepsilon}$; no 1.001-approximation in $(nm)^{1-\delta}$ [14]^{L}. *Status:* survives. Stated from SETH′: no $O^*((2-\delta)^N)$ CNF-SAT.

B8 Diameter: no $(3/2-\varepsilon)$-approximation in $O(m^{2-\varepsilon})$, undirected unweighted [70, Thm. 1]^{L}. *Status:* survives. Stated with conclusion "CNF-SAT in $O^*((2-\delta)^n)$". Later k-OV-based tradeoffs not checked here.

B9 Dynamic problems under SETH (#SSR, SubConn, SCC, ...) [7]^{L}. *Status:* survives^{I}. Stated via k-SAT with $O(n)$ clauses (sparsification), which is refuted; but the constructions use one node per clause, so the proofs run from CNF-SETH. Triangle, BMM and 3SUM-based bounds untouched.

B10 Results using $d = c \log n$ exactly, stated "under SETH". *Status:* premise refuted. They survive if restated under low-dimension OVC (A17).

B26 NSETH $\Rightarrow$ no deterministic fine-grained reduction from SAT to 3SUM, APSP [19]^{L}. *Status:* premise refuted. The barrier is lifted; no reduction follows.

Counting, quantum, circuits.

B17 #ETH bounds for the Tutte polynomial, the permanent and #IS [33, 25, 11]^{L}. *Status:* mixed. Conclusion refuted: #2-SAT and #IS on n variables/vertices at any density ([A], Corollaries 6, 7); #PM ([O], Corollary 1.3); sparse permanent, Boolean #CSP, the integer hyperbolas H_q (local encodings; also, independently and earlier, [10]). The line $T(x, 1)$ on sparse graphs, which we only conjecture under ($\mathrm{H_A}$), with identities checked numerically^{O}, is claimed by Wu and Wang [88, Cor. 10], assuming Theorem D.1 of [O]: all edge-cardinality counts of spanning forests. Premise refuted: the dense permanent and the rest. Under #SH, #2-SAT has no $O(2^{(n+m)^{1-\varepsilon}})$ algorithm [62, Cor. 5.1].

B18 Quantum-supremacy statements from ETH/SETH [63] and others. *Status:* mixed. Of 24 entries in [81, §6.2]: 5 refuted, 5 premise refuted, 1 moot, 1 open, 12 survive.

B27 ETH false $\Rightarrow \mathrm{E}^{\mathrm{NP}} \not\subseteq \mathrm{SIZE}(O(n))$ [89, Thm. 6.1]; see [46]^{L}. *Status:* fires. Section 4.1.

Rows are grouped by area rather than by number; the numbering follows our working map in [82]. Pătraşcu and Williams state row B3 as "Hypothesis 1 implies that CNF-SAT has an improved algorithm". Section 2

of [69] remarks that the earlier $f(k)n^{o(k)}$ bound for k-DomSet of Chen et al. concerns only k-SAT, which is precisely the distinction of Section 2.

4.1. **The circuit lower bound fires.** Williams [89, Theorem 6.1] proved that if ETH is false then E^{NP} has no circuits of linear size. Both manuscripts draw this consequence ([A]; [O], Corollary C.1), and [A] formalises it in Lean. Modulo the refutations, $\mathrm{E}^{\mathrm{NP}} \not\subseteq \mathrm{SIZE}(O(n))$ is therefore unconditional. OpenAI also state the quantitative form [68, Corollary 1.4]: some $L \in \mathrm{E}^{\mathrm{NP}}$ has no circuits of size $c\,n\sqrt{\log n}$ for infinitely many n, by Williams' argument and the local reductions of Jahanjou, Miles and Viola [46], applied to the constant-density case of their theorem. In a companion note [76] we prove the same statement, which we arrived at independently, from Anthropic's #3-SAT theorem, in the Turing-machine form of the clause-count bound of Imported Theorem 1.1 that its Lean development provides. We machine-check it in Lean on top of Anthropic's own circuit-lower-bound module (`enp_not_size_n_sqrt_log`; standard axioms; fresh kernel replay). The statement is OpenAI's; what we add is the formal proof. It holds only infinitely often, and the constant is not explicit. It gives nothing at $n \log n$ and nothing for NEXP. Wu and Wang [87] extend the bound to larger gate bases: $c\,m\sqrt{\log m}/\log\log m$ gates infinitely often over arbitrary-fan-in AND and OR with binary XOR, from the main theorem of [A], and $c\,m\sqrt{\log m}$ for fan-in-two circuits with a linear budget of arbitrary-fan-in XOR and symmetric gates, from an extension-ring form of Theorem D.1 of [O]. Carmosino et al. [19] note that disproving NSETH gives circuit lower bounds; for the k-CNF form refuted here this is subsumed by the above.

5. The new algorithms are few

The refutations give few explicit algorithms beyond k-SAT itself. In this section we list those we know of. We separate what needs only the formalised top-level statements from what needs internal results of [A] and what needs internal results of [O].

5.1. **From the main theorems alone.**

(i) k-SAT, #k-SAT, $\oplus k$-SAT and k-TAUT; #CNF-SAT of any width with $O(n)$ clauses (either engine), and with $n2^{o(L(n))}$ clauses ([A]; Proposition 2.1); linear-size Circuit-SAT.

(ii) [A], Corollary 7: for every $\delta > 0$, deterministic $O(2^{\delta n})$ algorithms that count the independent sets, cliques, vertex covers and dominating sets of each size s, and the proper k-colourings, of a graph on n vertices with any number of edges. Decision for IS, Clique, VC and k-Colouring was implied before by [44]. Counting independent sets and proper q-colourings is also [O], Corollary 1.3. Dominating Set and the counts by size are new in [A].

(iii) k-Hitting Set, k-Set Splitting, k-NAE-SAT, #2-SAT, sparse permanent, #PM, Boolean #CSP, and the integer Potts and flow hyperbolas of the Tutte plane on sparse graphs (local encodings, (F2)). #PM is [O], Corollary 1.3, for every graph; the sparse permanent, Boolean #CSP and the hyperbolas appear earlier in [10].

(iv) *Vertex Cover in* $2^{o(k)} + \mathrm{poly}(n)$ (obtained earlier by Bhattacharjee [10, Thm. 6.13]; we arrived at it independently). The LP kernel leaves at most $2k$ vertices in polynomial time [28, §2.5]. Corollary 7 then counts vertex covers of size k in $2^{2\delta k}$ for every δ. A single algorithm in time $2^{O(k \log\log k/\sqrt{\log k})} + \mathrm{poly}(n)$ follows by running the explicit n-bound of Imported Theorem 1.1 on the $O(k)$-variable 3-CNF of the proof of Corollary 7. This contradicts the conclusion of Cai–Juedes' ETH bound (B14), consistently with their own equivalence.

(v) *Sparse Max-Cut and Max-k-SAT, unweighted.* Bhattacharjee [10, Rem. 6.5] notes earlier that the circuit encoding in the proof of [O], Corollary 1.2, gives the exact optimum in this way; we arrived at it independently, and add the counts by value. Take $m = O(n)$ constraints with $O(1)$-bit weights. Encode each constraint by a forced indicator variable, add an $O(m)$-gate adder and a comparison with a target t, and apply Tseitin. The result is a 3-CNF with $O(n)$ variables whose models are the assignments of value exactly t. Corollary 6 counts them for each t, giving the exact optimum and the number of solutions of each value in $2^{\delta n}$. Weights up to W cost $O(m \log W)$ gates, so the main theorems do not cover polynomial weights. Bhattacharjee [10] obtains them by running the algorithm of [O] itself over truncated polynomial rings, a white-box extension.

(vi) M[1]-complete problems in FPT time (A15).

5.2. **From internal results of [A].** Our other results use three internal statements of [A] (a network-to-clique reduction, an algorithm for cliques of all levels, and the existence of its "pieces"), called ($\mathrm{H_A}$) in [75, 81]. They are not consequences of the top-level theorems. They are, however, theorems of [A]'s Lean development, and our Lean proofs are built on that development. So the results below that we call machine-checked use only the standard axioms (`propext`, `Classical.choice`, `Quot.sound`), have passed a fresh kernel replay, and stand or fall with [A]'s development.

(vii) *Sparse Hamiltonicity* [75]: the number of Hamiltonian cycles of a graph with n vertices and $O(n)$ edges in $O(2^{\delta n})$ for every δ, and in $2^{O(\max(m,n)/\sqrt{\log n})}$ in general; hence decision, search, Hamiltonian paths, sparse TSP with polynomial weights, directed cycles and ATSP. All are machine-checked in Lean, end to end on top of [A]'s development.

(viii) *Eulerian circuits* [78], a consequence of (vii). Counting Eulerian circuits is #P-complete [13]. For every $\delta > 0$ and Δ there is a machine that counts them on graphs of maximum degree at most Δ in time $(n+2)^C 2^{\delta m}$, and one machine does so for all graphs in $(n+2)^C 2^{Cm\log_2(\Delta+2)/\sqrt{\log_2(m+2)}}$. The reduction is Kotzig's bijection with the Hamiltonian cycles of the line graph of the double subdivision. Machine-checked; the formal object is Kotzig's transition-system definition.

(ix) *Exact quantum amplitudes* [81]: amplitudes of Clifford+T circuits with G two-qubit gates in $2^{O(G/\sqrt{\log G})}$, machine-checked. The polynomial-space variant and the marginal and sampling corollaries have written proofs only and are not claimed.

(x) *Deterministic factoring.* Rudich and Rousseau [72] used the counting engine to prove in Lean that one deterministic Turing machine computes the least prime factor of an n-bit integer within $C_k 2^{n/k}$ steps for every k, and conjectured the rate $2^{O(n/\sqrt{\log n})}$. Before 2026 every proven deterministic bound had a fixed positive exponent, the best being $N^{1/5+o(1)}$ [41]. The rate $2^{O(n/\sqrt{\log n})}$ was proved in Lean in a public repository [45] and on paper by Shunia [73], conditionally on two unrefereed preprints. We give an independent second Lean proof, along a different route [79]: one Turing machine outputs the least prime factor of N within $\lfloor C \cdot 2^{Cn/\sqrt{\log_2(n+2)}} \rfloor$ steps, by evaluating a two-dimensional Schönhage–Strassen multiplication network with Anthropic's counting theorem. Machine-checked.

(xi) *Class numbers and lattice-point counts* [77]: one Turing machine outputs, within $\lfloor C \cdot 2^{Cn/\sqrt{\log_2(n+2)}} \rfloor$ steps on n-bit inputs, the number of reduced forms of discriminant $-N$ (for fundamental discriminants, $h(-N)$), and others give $\sum_{m\le x}\tau(m)$, $\sum_{m\le x}\sigma(m)$ and the Gauss circle count. Before, the best proven deterministic bounds were fixed powers of the input. Machine-checked; Lean proves the count of reduced forms, whose equality with the class number is classical and not formalised. The same note machine-checks the least square root modulo m and the least quadratic non-residue modulo a prime, but these bounds are not new. Modulo a prime both take deterministic polynomial time, by OpenAI's proof of the quasi-Riemann hypothesis [66, Cor. 1.2] and by its deterministic factorisation of polynomials over $\mathbb{F}_p$ [67], both posted before the refutations; for composite m the least square root follows on paper from deterministic factoring [73], Hensel lifting and the Chinese remainder theorem. Vega [86] shows, unconditionally, that $h(-d)$ is computable in $\mathrm{FP}^{\Sigma_2^{\mathrm{P}}}$, a statement of a different kind.

(xii) *The parity of $\pi(x)$* [80]: $\pi(x)$ mod 2, and the parity of the number of primes in any interval $[A, B]$, in time $2^{O(n/\sqrt{\log n})} = x^{o(1)}$. Machine-checked. It gives neither prime finding nor the value of $\pi(x)$.

(xiii) *Further sparse problems, conjectured and not claimed* [82]: Steiner Tree, FVS, Connected VC and DS, Max-Leaf; the line $T(x,1)$ of the Tutte plane (forests counted by size) on sparse graphs, which is the line Brand, Dell and Roth [11] show #ETH-hard; sparse Max-Cut with polynomial weights. All are conditional on ($\mathrm{H_A}$). For the Tutte and connectivity problems the identities are checked numerically only; none is formalised. For FVS, Steiner trees and $T(x,1)$, Wu and Wang [88] give a different route, through [O] (Section 5.3). For Max-Cut and Max-k-SAT with polynomial weights, Bhattacharjee [10] gives an earlier, independent route. It replaces ($\mathrm{H_A}$) by the claim that the steps of [O] remain valid over a commutative $\mathbb{Q}$-algebra (unrefereed, not formalised).

5.3. **From internal results of [O].** Other follow-up works use statements inside [O], chiefly its Boolean tensor-contraction theorem (Theorem D.1). It proves the headline bound of [O], but OpenAI's Lean

development covers only the weaker bound of its Theorem 1.1, not Theorem D.1 [50, §2.3]. None of these works reports a formalisation of its own results; we list them as claimed.

(xiv) *Weighted counting.* Bhattacharjee [10] reproduces the algorithm of [O] and extends it to sums of local weights in any commutative $\mathbb{Q}$-algebra, with many weighted consequences, among them exact sparse Max-Cut and Max-k-SAT and the independence polynomial on every graph.

(xv) *Factoring.* Shunia [73], from the tensor-contraction theorem of [O]: the complete factorisation of an n-bit integer in deterministic time $2^{O(n/\sqrt{\log n})}$ and polynomial space (item (x) above).

(xvi) *Feedback vertex sets and forests.* Wu and Wang [88], assuming Theorem D.1: the number of feedback vertex sets of each size of any n-vertex graph, in time $\mathrm{poly}(n+m)2^{O(n(\log\log n)^2/\sqrt{\log n})}$ and polynomial space, and likewise induced trees and paths; and, with $s = n(1+\log_2(1+m/n))$, directed Hamiltonian cycles, spanning forests by number of edges and Steiner trees by number of vertices in $\mathrm{poly}(n+m)2^{O(s/\sqrt{\log s})}$.

(xvii) *Learning and inference.* Li [50], from a special case of Theorem D.1 over truncated polynomial rings: exact error profiles, hence exact empirical risk minimisation and exact sparse Max-CSP, and exact inference in graphical models of bounded degree and arity in $2^{O(n/\sqrt{\log n})}$.

5.4. **What gets no algorithm.** Nothing parameterised by treewidth or pathwidth gets an algorithm, except binary CSP on one class of bounded-degree graphs, with exponent $O(\mathrm{tw}/\sqrt{\log \mathrm{tw}})$ [64] (B20). Nor does any problem in P (edit distance, LCS, Fréchet, diameter, OV, 3SUM, APSP, k-Clique; for 3SUM and APSP see [6], which is unrelated to SAT), Set Cover by universe size, Subset Sum, dense Max-CSP, dense Hamiltonicity, or a planar problem (where $2^{O(\sqrt{n})}$ is already better).

6. Reading the map by area

In this section we read the two lists area by area: exact and parameterised algorithms, approximation, problems in P, and counting and quantum.

6.1. **Exact and parameterised algorithms.** The treewidth and pathwidth bounds (B1, B2) are the clearest case of a premise that survives because of how it is stated; this is the first half of Example 1.3. Lokshtanov, Marx and Saurabh build, from a CNF formula with n variables and m clauses, a graph of size polynomial in $n+m$ and treewidth roughly $n/\log_2 c$ (the precise form varies by problem). So a $(c-\varepsilon)^{\mathrm{tw}}$ algorithm would decide CNF-SAT in time $2^{(1-\delta)n}\,\mathrm{poly}(m)$. Nothing in the reduction requires bounded width or few clauses, and Remark 2.2 shows that the new algorithms reach only $m \le n2^{o(L(n))}$. To refute B1 one would need CNF-SAT with, say, n^2 clauses of width up to n in time $2^{(1-\delta)n}$. The width reduction would then need $k \approx \log n$, where the bound of Imported Theorem 1.1 is $2^{\Theta(n(\log n)^{3/2}\log\log n)}$, far above 2^n. The other half of Example 1.3, row B13, is contradicted outright. Corollary 7 of [A] is a reduction in the other direction: it encodes Independent Set on n vertices, at any density, as a 3-CNF with $O(n)$ variables.

The ETH-based bounds for parameterised problems (B19–B22) lose their premise, but are not contradicted: a $2^{\delta n}$ algorithm says nothing about $f(k)n^{o(k)}$ when $k \ll n$. The two exceptions are structural. Chen and Grohe's equivalence (A15) is an "if and only if", so its other direction now yields M[1] = FPT. And Vertex Cover parameterised by solution size has a linear kernel, so the parameter and the number of vertices coincide after preprocessing, and the $2^{\delta n}$ algorithm of [A] becomes a $2^{o(k)}$ algorithm (B14), as Bhattacharjee observed first. Whether W[1] $\neq$ FPT (A16) is unaffected: Chen and Grohe list its equivalence with ETH as open, and the refutations settle ETH without touching it. Most of the ETH-based bounds return in a weaker form under SH (A33): the transfer theorem of the MIT Hardness Group [62, Thm. 3.1] loses only ε in the exponent, and their Table 1 lists such bounds for planar, geometric and parameterised problems (B13, B14, B22).

6.2. **Approximation.** Gap-ETH is refuted by any exact $2^{o(n)}$ algorithm, so it falls with ETH; the gap does not help. The FPT-inapproximability results of Chalermsook et al. (B23) therefore lose their premise. So do the ETH-hardness results of Dinur–Manurangsi, Braverman–Ko–Weinstein and Aaronson–Impagliazzo–Moshkovitz (B24) and the ETH proof of PIH (B25). Two kinds of result survive. The first are the re-proofs under W[1] $\neq$ FPT: constant-factor k-Clique (Lin), $k^{1/H(k)}$-factor k-Clique (Karthik–Khot), $k^{o(1)}$-factor k-Clique (Chen–Feng–Laekhanukit–Liu) and $F(k)$-factor k-DomSet (Karthik–Laekhanukit–Manurangsi). Only the ETH-based versions, such as the $k^{o(1)}$ bound of Lin, Ren, Sun and Wang, lose their premise.

The second are results whose reductions are only mildly exponential, so that a weaker hypothesis suffices: Manurangsi's Densest-k-Subgraph bound, now stated under Almost-ETH (A33). The refutations give 3-SAT in $2^{n/(\log n)^c}$ for every constant $c < 1/2$. This is $2^{n^{1-o(1)}}$ but not $2^{n^{1-\gamma}}$, so Almost-ETH stands. We expect other quasi-polynomial-size reductions (Nash, free games, 2-CSP) to give weaker but non-trivial bounds under Almost-ETH; we have not checked this ([I]).

6.3. **Problems in P.** Here the map is almost entirely "survives". Edit distance, Fréchet distance and diameter are stated from CNF-SAT with poly(m) clauses (B5, B7, B8). LCS and DTW go through Most-OV or OV with dimension $d = m$ (B6), which CNF-SETH implies. What is lost is the derivation of the *low-dimension* OV conjecture ($d = c \log n$) from SETH, which uses sparsification to $O(n)$ clauses (A17, B10). OVC itself, 3SUM, APSP and the hyperclique hypothesis were never consequences of ETH. The dynamic lower bounds of Abboud and Vassilevska Williams (B9) are a borderline case. They are stated through sparsification, but by our reading proved by constructions that also run from CNF-SETH. The barrier of Carmosino et al. (B26), that NSETH forbids deterministic fine-grained reductions from SAT to 3SUM and APSP, disappears for k-SAT; no such reduction is known. Separately, Alman and Vassilevska Williams [6] claim truly subquadratic 3SUM and truly subcubic APSP by an algorithm for thin matrix products that has nothing to do with SAT. This changes no status in the map, which records only what follows from the ETH refutations (A23–A25).

6.4. **Counting and quantum.** Counting statements behave like decision ones. #ETH and the k-CNF form of #SETH fall (A6, A7). Lower bounds proved by local reductions on sparse instances are contradicted by explicit algorithms (B17): Corollaries 6 and 7 of [A] count independent sets and models of 2-CNFs at any density, and local encodings handle sparse permanents, perfect matchings and the integer hyperbolas of the Tutte plane. The non-local cases need internal results: Hamiltonian cycles are machine-checked under ($\mathrm{H_A}$) (B16), while forests, the line $T(x, 1)$, are conjectured under ($\mathrm{H_A}$) and claimed by Wu and Wang under Theorem D.1 of [O] (B17). Dense permanents, and the Tutte plane on dense graphs, are untouched. For the quantum statements (B18) the same split appears: sparse Clifford+T circuits fall under ($\mathrm{H_A}$), while dense or average-case conjectures survive [81, §6].

7. Related work

We organise related work by idea: collections of ETH-based lower bounds, the structure of SETH and its variants, and four kinds of follow-up work: replacement hypotheses, new algorithms, constants and barriers, and expositions.

Collections of lower bounds. Lokshtanov, Marx and Saurabh [55] survey lower bounds based on ETH, and so does Chapter 14 of the textbook of Cygan et al. [28]. Both were written to *use* the hypotheses. In contrast, we take each bound as an object and ask which of its premises the refutations remove.

The structure of the hypotheses. Closest to our work are Section 1.1.3 of OpenAI's manuscript [68], which already separates fixed from unbounded width for the Fréchet, diameter and treewidth bounds and points to OVC for LCS, and the ledger of Li [50, §6], posted on 29 October 2026, which takes stock of learning theory. Li's table of fine-grained hypotheses agrees with ours where they overlap: the k-CNF forms of ETH and SETH, Gap-ETH and the sparse Max-SAT hypotheses are false; CNF-SETH and its counting form are open; the hard instances of CNF-SETH need superlinearly many clauses; and low-dimension OV no longer follows from a hypothesis about SAT [50, Table 1, Prop. 6.1]. His Lemma 3.5 and Corollary 3.6 sharpen Remark 2.2(c): under CNF-SETH, width $O(\log n)$ already needs time $2^{(1-\gamma)n}$. In contrast, we classify each row, including the approximation, parameterised, counting and fine-grained bounds outside learning, against the premise its proof uses. Cygan et al. [27] map which problems are equivalent to SETH and which are tied to the Set Cover Conjecture instead. Calabro, Impagliazzo and Paturi [18] relate clause width to clause density, and Gao, Impagliazzo, Kolokolova and Williams [38] separate low- and moderate-dimension OV. Carmosino et al. [19] introduce the nondeterministic variants. These works draw the distinctions our map depends on. In contrast, they ask which hypotheses imply which; we ask what is left once the k-CNF forms are false. The width–density relation of [18] becomes the dividing line of Section 2.

Replacement hypotheses. Manurangsi [59] restates his Densest-k-Subgraph bound under Almost-ETH. The MIT Hardness Group [62] prove that Almost-ETH, which they call Weak ETH, is equivalent to the Satisfiability Hypothesis of Stearns and Hunt, and #SH to Weak #ETH; their transfer theorem keeps the usual reductions at the cost of ε in the exponent. Li [50, Hyp. 2.14] proposes a length form of ETH at an explicit rate. These

works ask which hypothesis should carry lower bounds from now on. In contrast, we ask what the refutations do to the hypotheses and theorems already in the literature; rows A33, B13, B14, B22 and B24 use their results.

New algorithms. Bhattacharjee [10] extends the algorithm of [O] to weighted sums, Nguyen [64] decides binary CSP with a sublinear treewidth exponent on one class of graphs, Rudich and Rousseau [72] and Shunia [73] factor, Wu and Wang count feedback vertex sets [88] and extend the circuit lower bound to larger gate bases [87], and Li [50] speeds up learning of local hypothesis classes. Our own notes count Hamiltonian cycles [75], Eulerian circuits [78] and quantum amplitudes [81], compute class numbers and lattice-point counts [77] and the parity of $\pi(x)$ [80], factor [79], and machine-check OpenAI's superlinear circuit lower bound [76]; [81, §6.2] classifies the fine-grained quantum-supremacy conjectures, which row B18 summarises. Two neighbouring problems were settled before the refutations, by other results of OpenAI: square roots and least quadratic non-residues modulo a prime take polynomial time [66, 67] (Section 5). Vega's placement of class numbers in $\mathrm{FP}^{\Sigma_2^{\mathrm{P}}}$ [86] is unconditional. These works add algorithms one problem at a time. In contrast, this note asks what the refutations do to the hypotheses and lower bounds themselves.

Constants, crossover points and barriers. OpenAI's manuscript states no constants [32]. Szeider [74] computes from OpenAI's own parameter choices that its main-text algorithm is an exhaustive search on every input with fewer than $2^{10^{10}}$ variables, and its Appendix D algorithm on every input with fewer than 2^{131071} variables and clauses; an addendum treats [A]. Majdoddin [58] makes the recursion of [A] explicit and lowers its crossover points by proved changes: at clause density one, to about 2^{420} clauses from 2^{33858}, and for general 3-SAT to an estimated $2^{8.6\cdot 10^9}$ variables. He also proves a barrier: every grid algorithm in a rewrite-and-merge class that contains the recursion has exponent at least $1/\log_2 K$ per edge bit on clique sums with K nodes, so in his main-term model none gains for general 3-SAT below about 2^{6000} variables [58, Thm. 4.13, Cor. 4.15, Cor. 4.19]. Dias and Lima [32] extract the constants of OpenAI's tensor-contraction proof, whose certified time at bounded density stays above 2^n unless $\log_2(n+m) > 10^9$, and prove limits in a cost model for its interpolation scheme, such as $2^{\Omega(\sqrt{\mathrm{tw}})}$ at treewidth tw. In contrast, the map concerns asymptotic statements only; these works explain why nothing in it is practical.

Expositions. O'Donnell [65] and Kominers [47] expound the algorithm of [O], the second in a qualitative form. In contrast, we use both manuscripts only through their stated theorems.

8. Summary: three findings

Table 1 gives the number of rows per status.

Status	Hypotheses (34)	Theorems (27)	Total
refuted	13	–	13
conclusion refuted	–	3 (B13–B15)	3
fires	–	1 (B27)	1
premise refuted	–	10	10
survives	20	9	29
mixed	1 (A29)	4 (B16–B18, B24)	5

Table 1. Counts per status. Refuted: A1–A3, A5–A7, A9, A11–A13, A15, A22, A34. Premise refuted: B4, B10, B12, B19–B23, B25, B26. Survives: A4, A8, A10, A14, A16–A21, A23–A28, A30–A33, B1–B3, B5–B9, B11.

Three points stand out. First, the most-cited SETH lower bounds survive with an intact premise: those for treewidth DP and for edit distance, Fréchet distance, LCS and diameter. They are proved from CNF-SAT with polynomially many clauses. What dies is "k-SETH ⇒ low-dimension OVC" and anything that genuinely needs sparsification. Second, Gap-ETH is refuted. So the Gap-ETH programme in parameterised inapproximability and the ETH proof of PIH lose their premises; the parts re-proved under W[1] ≠ FPT survive. Third, the frontier of the new algorithms in clause density is sharp in an unusual way: $n \cdot 2^{o(L(n))}$ clauses through the explicit rate of [A], constant density through [O], and nothing at $n^{1+\Omega(1)}$.

9. Limitations

- *The map stands or falls with the claimed refutations* [24, 68], whose expert review is pending. If they fail, every "refuted" becomes "open" and every "premise refuted" reverts to a valid conditional theorem; Proposition 2.1 and the observations of Section 5 lose their proofs. Results marked (H_A) use internal statements of [A] that are not consequences of its top-level theorems. They are theorems of [A]'s Lean development, so these results stand or fall with that development.
- *A refutation removes premises.* Except through (F2) and the network theorems, it gives no algorithm, and "premise refuted" is never evidence that a conclusion is false.
- *The sources were checked, but the map is not exhaustive.* Every row was checked against its primary source; the nine rows that earlier rested on secondary sources were re-verified for this note (Section 1.5), and no row is now secondary. Rows marked ^{F} have no single primary statement. Rows marked ^{I} are our inferences and are argued only in the reason column. B8 omits later diameter tradeoffs. [A] reports more than 800 papers with results proved only under ETH or a hypothesis implying it. A novelty search of arXiv and Hexagon on 5 November 2026 added row A34, the notes in A23–A25 and the follow-up works of Section 7; the literature is moving fast, and we may have missed others.
- *Our written arguments are not machine-checked.* Proposition 2.1 and the observations (iv)–(v) of Section 5 are short written arguments, produced with AI assistance and not machine-checked or reviewed by a human expert. Only the results so labelled in Section 5 are machine-checked, on top of the labs' Lean developments. No human has yet checked that their formal statements match the printed ones.
- *Nothing is practical.* [A] notes that its bound does not beat 2^n for 3-CNFs with $5n$ clauses unless $n > 10^{10\,000}$. Later computations lower such crossover points but leave them astronomical: about 2^{420} clauses at density one for [A] [58], while the Appendix D algorithm of [O] is an exhaustive search below 2^{131071} variables and clauses [74]. All algorithms here are galactic.

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